

# Use Case Formats in Software Engineering

## 1 Introduction

Use cases can be documented at varying levels of detail depending on the stage of development and the purpose of analysis. There are three common formats for describing use cases in software engineering:

1. **Brief Format**
2. **Casual Format**
3. **Fully Dressed Format**

### Key Idea

Each format represents a different level of formality. Early stages of requirement gathering may use brief or casual formats, while the fully dressed format is used for detailed system design and testing.

## 2 1. Brief Format

The **Brief Use Case** format provides a short, one-paragraph summary describing the main actor, goal, and essential outcome.

### Definition

A **Brief Use Case** is a high-level summary that captures the essence of the interaction between actor and system in just a few sentences.

### When to Use:

- During early requirement elicitation.
- When prioritizing or identifying potential use cases.
- For executive or stakeholder overviews.

### Example – ATM Withdrawal (Brief):

A customer inserts an ATM card, enters the PIN, and requests cash withdrawal. The system verifies credentials, dispenses cash, and updates the account balance.

This level of detail helps stakeholders quickly understand what the system should accomplish without focusing on implementation details.

## 3 2. Casual Format

The **Casual Use Case** format provides a few structured sections, typically written in paragraphs, describing main flows and alternate flows in moderate detail.

### Definition

A **Casual Use Case** is a semi-structured narrative describing the actor's goal, main success scenario, and possible variations, without strict formalization.

### When to Use:

- When the system is moderately complex.
- For communication between analysts and stakeholders.
- Before moving to detailed modeling.

### Example – Online Food Ordering (Casual):

**Actor:** Customer

**Goal:** Order food from an online restaurant.

#### Main Flow:

1. Customer logs in to the application.
2. Browses restaurant menu and selects items.
3. Confirms the order and makes payment.
4. System sends confirmation and estimated delivery time.

#### Alternative Flow:

1. If payment fails, system prompts the user to retry with a different method.

The casual format captures most real-world scenarios without requiring complete formal structure.

## 4 3. Fully Dressed Format

The **Fully Dressed Use Case** is the most detailed and formal representation. It includes all essential elements such as preconditions, triggers, success and failure paths,

and postconditions.

#### Definition

A **Fully Dressed Use Case** provides a comprehensive and structured description of user-system interaction, suitable for detailed system design, implementation, and testing.

#### When to Use:

- During detailed requirement analysis or system design.
- For mission-critical or complex systems.
- As a foundation for creating test cases or documentation.

**Example – Online Shopping Checkout (Fully Dressed):**

**Use Case Name:** Checkout Process

**Actor:** Registered Customer

**Goal:** Complete an online purchase.

**Scope:** E-commerce Web Application

**Level:** User Goal

**Precondition:** Customer has items in the cart and is logged in.

**Trigger:** Customer selects “Proceed to Checkout.”

**Main Success Scenario:**

1. System displays checkout page with order summary.
2. Customer enters shipping and payment details.
3. System verifies payment and confirms the order.
4. System sends order confirmation email and displays a success message.

**Extensions (Alternate Flows):**

1. If payment verification fails, system prompts for another payment method.
2. If the item is out of stock, system notifies the customer and removes it from the cart.

**Postconditions:**

- Order is recorded in the database.
- Confirmation email is sent to the user.

This level of documentation provides clarity and traceability throughout the SDLC, making it ideal for large-scale projects.

## 5 Comparison of Formats

| Feature        | Brief                 | Casual                    | Fully Dressed                |
|----------------|-----------------------|---------------------------|------------------------------|
| Structure      | One paragraph summary | Informal narrative        | Formal and detailed template |
| Purpose        | Quick overview        | Requirement clarification | System design and testing    |
| Detail Level   | Low                   | Moderate                  | High                         |
| Ideal For      | Brainstorming         | Requirement validation    | Implementation phase         |
| Example Length | 3–5 lines             | Few paragraphs            | Full page                    |

## 6 Summary

Different projects require different levels of detail in use case documentation. **Brief** use cases support early brainstorming, **Casual** use cases clarify requirements, and **Fully Dressed** use cases ensure design precision and traceability.

*Choosing the right use case format ensures clarity, saves effort, and aligns stakeholders at every stage of software development.*